How Young People are Using ChatGPT for Social Support

ANONYMOUS AUTHOR(S)

General-Purpose Conversational Companions (GPCCs) such as ChatGPT have rapidly become part of emerging adults' everyday lives. Yet despite growing research attention on conversational companions for social support, we lack systematic documentation of what support young adults actually seek from GPCCs and how these systems respond. In this study, we collected in-the-wild ChatGPT usage histories from 85 emerging adults (ages 18–25), comprising approximately 157,000 messages exchanged over three years across 17 countries. Through mixed-methods analysis combining reflexive thematic analysis with computational text analysis, we identified ten distinct usage patterns across four life domains: Homework & Learning, Self-Sufficiency, Health & Well-Being, and Relationships. We further characterized five ChatGPT response patterns: Task Takeover, Scaffolding, Emotional Attunement, Expert Knowledge, and Supplying New Ideas. We synthesize these findings into a design space and argue that developmentally supportive GPCCs should shift toward scaffolding user action.

CCS Concepts: • **Human-centered computing → Empirical studies in HCI**; **HCI theory, concepts and models**.

Additional Key Words and Phrases: Self-disclosure; AI Companions; Emerging Adulthood; Development; Longitudinal Analysis; ChatGPT

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1 INTRODUCTION

Social support—defined as the provision of assistance or comfort by family, friends, neighbors, colleagues, caregivers to help cope with biological, psychological, and social stressors [23, 25]—is a cornerstone of the health, well-being, and development of young people in their emerging adulthood[16]. Traditionally, parents and peers have served as the primary sources of social support for emerging adults (18-25) navigating life's challenges [34]. Yet increasingly, young people are turning to a new actor: General-Purpose Conversational Companions (GPCCs) such as ChatGPT. On November 30, 2022, ChatGPT was released to the public, and by the end of the year, 30% of young adults in the United States were using it for homework support[66, 67]. By 2025, 58% of all messages sent to ChatGPT came from young adults under 25[21]. Today, ChatGPT has become an essential part of young people's everyday lives.

Research on conversational companions (CC) as sources of social support has gained significant attention within the HCI community, particularly in the context of young people [9, 47, 51, 75, 83]. Studies have documented that emerging adults engage with CCs across all dimensions of social support—appraisal, informational, emotional, and instrumental [9, 75]. They perceive CCs as non-judgmental safe spaces [24, 41], and, in many cases, prefer CC responses over those of human peers and trained counselors [83]. They form meaningful relationships with CCs, reporting feelings of social connection and reduced loneliness[52, 54, 68, 69, 82].

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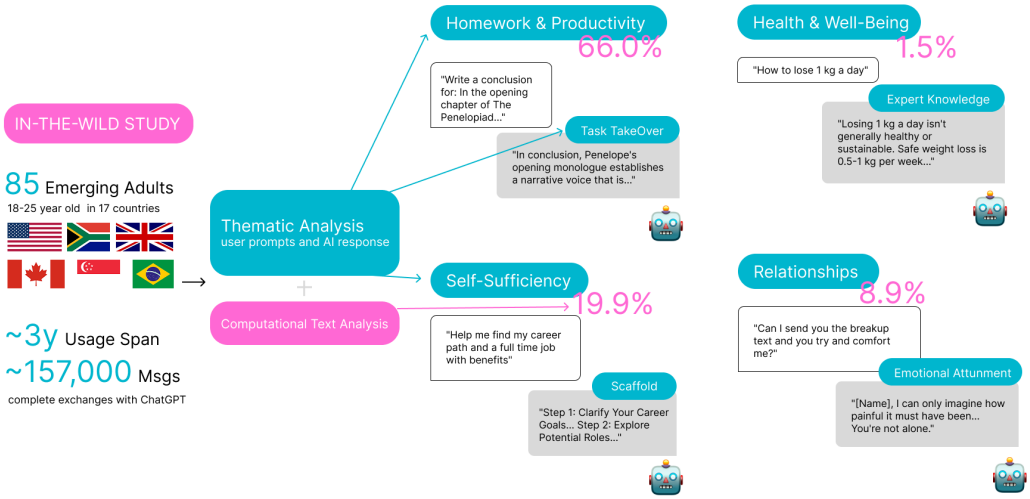


Fig. 1. Overview of our mixed-methods analysis of 85 emerging adults' (ages 18-25) in-the-wild ChatGPT usage across 17 countries over approximately three years (157,000 messages). Through thematic analysis and computational text analysis, we identified ten user usage patterns spanning Homework & Productivity (66.6%), Self-Sufficiency (17.6%), Health & Well-Being (10.5%), and Relationships (5.4%). We also characterized five ChatGPT response patterns—Task Takeover, Scaffolding, Expert Knowledge, Emotional Attunement, and Supplying New Ideas.

Despite this growing body of work, our understanding of how young adults seek and receive social support through CCs is still developing. Our understanding is currently limited in three important ways. First, the majority of existing research has examined CCs such as Replika[51], even though GPCCs now dominate young people's usage. GPCCs are conversational companions powered by LLMs with a wide range of capabilities—from answering homework questions[49] to mental health support[39]—and include platforms such as ChatGPT (OpenAI), Gemini (Google), and Claude (Anthropic).

Second, most studies relied on single-shot experiments despite evidence that long-term usage with CCs is the key mechanism for supportive relationship formation with CCs [51, 70]. GPCCs uniquely fill this gap as many young people engage with them consistently over months to years due to their general-purpose utility [66, 67]. Long-term usage of GPCCs raises important questions about the kinds of support relationships that develop over time and how they may differ from those formed with purpose-built CCs [19, 70].

Third, our field's understanding of social support is built largely on what users choose to share publicly. Existing studies have heavily used self-report data—screenshots on Reddit[51, 84] and app store reviews[75] that are subject to selection bias and offer only fragmentary glimpses of actual exchanges [9, 24, 51, 52]. Understanding the full breadth of social support requires analysis of in-the-wild conversation logs to assess the types of requests young people make and the responses–including the routine, unremarkable interactions that users would not think to post online but that matter for understanding how support unfolds.

Lastly, understanding how ChatGPT responds to young people's support-seeking is also an urgent societal concern. In November 2025, the American Psychological Association issued a health advisory warning that GPCCs "can have unintended effects and even harm mental health" [3]. A RAND study published the same month found that 13.1% of U.S. adolescents and young adults—approximately 5.4 million individuals—now use CCs for mental health advice [45]. Legislators have also accelerated regulatory efforts. The U.S. Senate held hearings on "Examining the Harm of AI Chatbots" [4], and states such as New York, California, and Utah have enacted or proposed legislation requiring safety protocols for conversational companions. Despite this urgency, public debates have proceeded largely without systematic documentation of how CCs respond to young people's requests.

These gaps motivate the present study to examine ChatGPT, the most widely used GPCC by young people [21]. We pose the following research questions:

(1) **RQ1:** What types of social support are emerging adults seeking from ChatGPT?
(2) **RQ2:** How does ChatGPT respond to emerging adults' support-seeking requests?

To address these questions, we collected naturalistic ChatGPT usage histories from 85 emerging adults, comprising approximately 157,000 messages exchanged over three years. We conducted mixed-methods analysis combining reflexive thematic analysis with computational text analysis. Through characterization of user requests and ChatGPT responses, we identified ten distinct usage patterns across four life domains (Homework & Learning, Self-Sufficiency, Health & Well-Being, and Relationships) and five GPCC response patterns (Task Takeover, Scaffolding, Emotional Attunement, Expert Knowledge, and Supplying New Ideas). We synthesize these findings into a design space for developmentally supportive GPCCs, defined by two dimensions: the level of tangible support (from Task Takeover to Task Scaffolding) and the level of emotional support (from impersonal to personal contexts).

This research makes the following contributions:

- **A taxonomy of young adult social support patterns.** We provide the first empirical account of in-the-wild ChatGPT usage among emerging adults, documenting how social support extends beyond homework into health, relationships, and identity domains traditionally served by human supporters.
- **A taxonomy of ChatGPT response patterns.** We identify five systematic response patterns, mapping informational, emotional, and tangible support dimensions onto observed GPCC behaviors.
- **A design space for developmentally supportive GPCCs.** We propose a two-dimensional framework that distinguishes the level of tangible support (Task Takeover vs. Task Scaffolding) from the level of emotional support (impersonal vs. personal), and argue that developmentally supportive GPCCs should shift toward scaffolding user action.
- **Tangible support as the relationship driver**. We extend social support theory to the GPCC context by identifying tangible support—historically peripheral in CC research—as the mechanism through which young adults form sustained relationships with GPCCs, creating pathways from instrumental use to emotional disclosure.
- **A methodology for naturalistic GPCC research.** We demonstrate a replicable approach for collecting and analyzing in-the-wild GPCC conversation data, addressing a key methodological gap in prior work that relied on self-report and publicly shared content.

Given the unprecedented ability of GPCCs to deliver social support at any time and in any place, understanding young people's usage patterns in naturalistic contexts is both important and urgent. That GPCCs can provide support across every life domain does not mean they should do so without constraint. How support is prompted and delivered has direct

implications for the design of GPCCs, and whether these systems ultimately scaffold or supplant the developmental processes of emerging adulthood carries significant consequences for young people's well-being in a future where GPCCs are here to stay.

2 RELATED WORK

2.1 Social Support During Emerging Adulthood

Social support refers to the provision of assistance or comfort to help cope with biological, psychological, and social stressors by family, friends, neighbors, colleagues, caregivers[36, 78]. It is a multidimensional construct that has been consistently linked to positive health outcomes, psychological well-being, and resilience in the face of stress[23]. House and Weiss identified four types of social support: emotional support (e.g., expressions of empathy, caring, and reassurance), informational support (e.g., advice and relevant information), instrumental support (e.g., tangible and practical assistance), and appraisal support (e.g., feedback useful for self-evaluation) [36, 78]. Subsequent research has often consolidated these into three core dimensions—informational, emotional, and tangible support—which we adopt in this paper[71].

The effectiveness of social support depends not only on its availability but on its *fit* with the stressor at hand [23]. Cohen and Wills identified two mechanisms: a main effect model, in which social integration broadly benefits well-being, and a stress-buffering model, in which support protects well-being specifically in the presence of stressful events [17, 18, 20]. Cutrona and Russell's [25] optimal matching model further specifies that controllable stressors benefit most from informational and tangible support, while uncontrollable stressors benefit more from emotional support. This matching principle has important implications for CC-delivered social support, as conversational agents must somehow discern the nature of users' needs to respond appropriately.

A further nuance is the distinction between perceived support—the belief that help would be available if needed—and received support, the actual provision of supportive behaviors [20]. Perceived support consistently shows stronger associations with well-being than received support [40, 80]. This distinction becomes especially relevant in the context of CC-delivered support, where young people may receive extensive assistance yet not perceive themselves as genuinely supported.

For emerging adults (18-25) navigating the transition to adulthood, social support plays a particularly critical role[10, 16, 42, 48, 57, 60, 62, 64]. Over the past half-century, socioeconomic shifts in industrialized nations have elongated the transition to adulthood[5, 65]. This has given rise to a distinct developmental period–emerging adulthood–marked by identity exploration, instability, and the gradual assumption of adult responsibilities[6–8]. Meta-analytic evidence demonstrates that associations between social support and well-being are especially strong during this period [16, 48, 60, 64]. Traditionally, parents and peers have served as the primary sources of support [34], but young people often hesitate to seek help from these sources due to concerns about judgment, burdening others, or revealing vulnerabilities [59]. This reluctance creates an opening for alternative support sources: conversational companions (CCs).

2.2 Conversational Companions as Sources of Social Support

The HCI community has long examined conversational companions (CCs) as potential sources of social support from early rule-based agents systems like ELIZA[79] through retrieval-based agents such as Woebot [28], Wysa [29], and LLM-based agents such as Therabot [33], Replika [41, 51, 53, 69, 75, 84] and ChatGPT [39, 68, 73]. Several features make CCs

uniquely appealing to young people. CCs offer perceived 'safe spaces', free from peer evaluation [12, 44, 75]. Young adults turn to Replika and ChatGPT when they feel lonely [43], need privacy [39, 73], or seek identity exploration [44]. CCs' constant availability further attracts young adults, particularly those who need immediate support [39] or experience barriers to social connection [12] and mental health care [73]. CCs are also highly patient [39, 69]. Design features such as reciprocal self-disclosure [51] and conversational memory further enhance these affordances [38, 41, 47, 68].

Empirical research documents that young people engage with CCs across all dimensions of social support. Brandtzaeg et al. [12] found that young people used Woebot for appraisal , informational, emotional, and instrumental support. Similarly, Ta et al. documented that Replika users sought both emotional support (companionship, reduced loneliness) and informational support (advice on daily problems)[76]. Users form meaningful relationships with CCs [41, 53, 69, 82], and in some cases prefer CC responses over those of human peers and trained counselors [83]. Pan and de Graaf argue the human–chatbot support relationship is bidirectional, identifying limited reciprocal exchange—such as users teaching chatbots—as the key relational dynamic[51]. Skjuve et al.'s [70] longitudinal research further demonstrated that sustained relationships with chatbot companions develop over time, though sustaining engagement requires ongoing perceived value that companionship chatbots may lack.

Despite these findings, the evidence base faces important constraints. Only 10.4% of CUI studies involve naturalistic settings [50]. The majority of CC research relies on self-report data—screenshots on Reddit[41, 51, 84], app store reviews[75], and retrospective interviews[9, 22, 39, 52, 69, 73]—that are subject to selection bias and offer only fragmentary glimpses of actual exchanges. As a result, empirical studies have yielded different findings depending on participants selection. Studies that sample current CC users generally reach positive conclusions [70], whereas studies that assign non-users to interact with CCs find that participants lose interest quickly [24]. Most critically, existing research has predominantly examined purpose-built CCs such as Replika [51] and Woebot [9], even as general-purpose conversational companions now dominate young people's usage.

2.3 General-Purpose Conversational Companions

The commercialization of General-Purpose Conversational Companions (GPCCs)—systems powered by LLMs with broad task capabilities, including ChatGPT (OpenAI)[74], Gemini (Google)[2], and Claude (Anthropic)[1]—introduced a paradigm shift in how young people access support. Unlike purpose-built therapy or companion-focused platforms, GPCCs were not designed for social support but have been widely adopted for this purpose [35, 39, 54, 68].

A critical distinction between GPCCs (e.g., ChatGPT) and CCs (e.g., Replika) is how different types of social support emerge during use. Song et al. [73] found that users were often not seeking an emotional support tool when they first turned to GPCCs for support—rather, they were already using them for other tasks and, finding themselves in distress, turned to the resource already at hand. Jung et al. reported a similar pattern among users who adopted ChatGPT for mental health support [39]. This pathway—from instrumental use to emotional disclosure—suggests that understanding GPCC social support requires examining the full breadth of usage, not only emotionally salient conversations.

Understanding how GPCCs respond to young people's support-seeking is an urgent societal concern. By July 2025, ChatGPT alone had reached 700 million users sending 18 billion messages per week[66], with 58% of all messages coming from young adults under 25[21]. A RAND study found that 13.1% of U.S. adolescents and young adults—approximately 5.4 million individuals—now use CCs for mental health advice[46]. Users have reported that ChatGPT lacks competence as a therapist and provides inaccurate mental health information [39]. Trained psychologists rated commercial chatbots as adequate on only 31% of therapeutic-approach items and 39% of risk-monitoring items [72]. The American Psychological Association's 2025 health advisory warned that GPCCs "can have unintended effects and even harm mental health" [3],

and OpenAI's own research found that heavier platform users were statistically significantly lonelier and socialized less [27, 54]. Legislators have accelerated regulatory efforts, with the U.S. Senate holding hearings on GPCC harms [4] and states enacting safety legislation. Despite this urgency, public debates have proceeded largely without systematic documentation of how GPCCs actually respond to young people's everyday requests.

These research gaps motivate the present study. We examine ChatGPT, the most widely used GPCC by emerging adults (18-25) [66], through direct analysis of naturalistic conversation logs to address two questions: (1) What types of social support are young adults seeking from ChatGPT? (2) How does ChatGPT respond to young adults' support-seeking requests?

3 METHOD

This study employed a mixed-methods approach. We collected and analyzed a corpus of naturalistic conversation logs to understand how emerging adults interact with ChatGPT. The data collection and analysis procedures described below were approved by the first author's Institutional Review Board (IRB) and are part of a larger project on teens' and young adults' CC usage.

3.1 In-The-Wild Emerging Adult ChatGPT Usage Collection

Unlike prior studies that relied on screenshots and Reddit posts to understand human–CA interactions [9, 51, 52, 75], we collected young adults' complete ChatGPT usage histories exported directly from OpenAI's platform. This approach captures the full breadth and context of support-seeking behaviors as they naturally occur, rather than as retrospectively reconstructed.

We recruited emerging adults through the Prolific academic research platform between November 2024 and May 2025. We targeted individuals aged 18 and 25, the developmentally sensitive period of emerging adulthood[5, 6]. Prolific's verification system was used to confirm participant age. We recruited internationally to ensure our findings were not limited to a Western context, as research has shown that supportive CCs often lack cultural relevance for non-Western users [39, 63].

We screened for sufficient ChatGPT engagement (i.e., requiring that participants self-reported using ChatGPT more than ten times in the preceding two weeks). After providing informed consent, participants used OpenAI's standard data export function to download their ChatGPT conversation history and shared the exported files with our research team.

We received initial submissions from 510 respondents. To ensure analytic depth and linguistic consistency, we applied two inclusion criteria: (1) conversations were conducted primarily in English, and (2) the exported log exceeded ten pages of text for meaningful analysis. This screening yielded a final dataset of 85 participants (see Table 1 for demographics) whose logs contained 157,188 messages spanning up to three years of ChatGPT use. The majority of participants (64.7%) had engaged with ChatGPT regularly for more than one year, with an average usage span of 423 days (Mdn = 431, SD = 232, range: 5–819 days).

3.2 Method of Analysis

To answer our research questions at multiple levels of analysis, our mixed-methods approach proceeded through three complementary phases:

Table 1. Participant Demographics N=85

Characteristic			Characteristic		
Age			**Gender**		
Mean (SD)	21.7 (2.0) years		Male	43	53.8%
Median	21.0 years		Female	37	46.2%
Range	18–25 years				
Ethnicity			**Primary Language**		
White	31	39.2%	English	62	77.5%
Black	22	27.8%	Spanish	3	3.8%
Asian	13	16.5%	Chinese	3	3.8%
Mixed	11	13.9%	German	3	3.8%
Other	2	2.5%	Polish	2	2.5%
			Other languages[a]	6	7.6%
Country of Residence					
United States	37	46.2%			
South Africa	15	18.8%			
United Kingdom	8	10.0%			
Canada	6	7.5%			
Other countries[b]	13	17.5%			

[a]Other languages include Afrikaans (2), Italian (2), Armenian (1), French (1), and Other (1).
[b]Other countries include Mexico (3), Germany (3), Poland (2), Australia (2), Italy (2), New Zealand (1), and 1 other.

(1) In the exploration phase, we calculated descriptive statistics on the full dataset of 157,188 messages from all 85 participants to unpack patterns of longitudinal usage. We computed engagement metrics such as interaction span (days between first and last message), message volume per participant, conversation counts, and message length distributions for both user prompts and ChatGPT responses.
(2) Then, two authors performed reflexive thematic analysis on chatlogs to characterize user prompts (RQ1) and ChatGPT response patterns (RQ2).
(3) Finally, we conducted frequency and co-occurrence analyses on the coded data to quantify prevalence and distribution of usage types and CC response patterns.

3.3 Reflexive Thematic Analysis

We conducted reflexive thematic analysis, an approach that emphasizes the researcher's active role in knowledge production and interpretive depth (in contrast to post-positivist approaches that rely on inter-coder agreement)[13–15]. The process was iterative and multi-staged, proceeding through four stages:

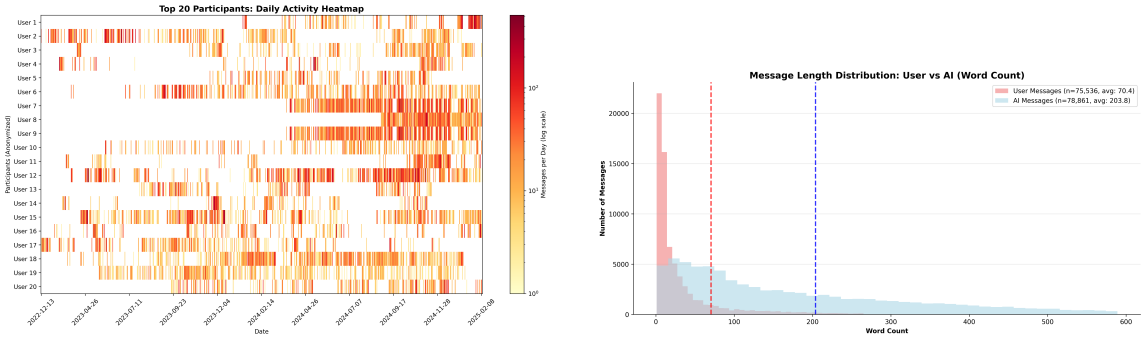
(1) In stage 1, the first author immersed themselves in the data by reading all conversation logs multiple times to develop initial inductive codes that captured salient usage patterns.
(2) In stage 2, we applied and refined these codes on a random sample of 100 conversations stratified by message length, producing ten distinct usage categories answering RQ1.
(3) In stage 3, we shifted to ChatGPT responses. We developed initial codes inductively from the same 100-conversation sample. We stratified this analysis by user usage categories to capture how ChatGPT behavior shifted across contexts (e.g., homework help versus emotional support).

(4) In stage 4, we applied finalized codebooks for both user usage and ChatGPT responses to an additional 150 previously unseen conversations, yielding a total annotated sample of 250 conversations stratified by message length(2.6% of total).

4 RESULTS PART 1: WHAT TYPES OF SOCIAL SUPPORT ARE YOUNG ADULTS SEEKING FROM CHATGPT?

4.1 Descriptive Statistics

In total, 85 participants exchanged 157,188 messages with ChatGPT. The majority (64.7%) had engaged with ChatGPT regularly for more than a year. Average total usage was 423 days (mdn=431.0, SD=232, min=5, max=819 days). Participants initiated an average of 107 conversations (mdn=71) with ChatGPT, during which they exchanged an average of 1,859 messages per person (mdn=895, SD=2416, min=30, max=13,178). This distribution was notably right-skewed, with 13% of participants (N=11) sending over 4,000 messages each. The majority of participants (71%, N=60) exchanged fewer than 600 total messages per person. Conversations were asymmetrical and reflected a pattern in which users posed concise queries and ChatGPT responded elaborately. Users' prompts averaged 19 words, and 85% contained no more than 20 words. In contrast, ChatGPT's responses were six times longer and averaged 123 words. Usage statistics are shown in Figures 2.



(a) Daily activity heatmap of top 20 participants

(b) Full word count distribution

Fig. 2. Usage Pattern showing (a) highly heterogeneous daily activity intensity across time for our participants and (b) User vs CC message length distributions showing asymmetric communication patterns: users average 19.4 words while CC responses average 122.8 words

Participants queried ChatGPT about a wide range of topics. Many of these conversations reflected participants' transitional life stage, in which they were tasked with mastering academic concepts, taking responsibility for their domestic life, and planning and launching their careers. We clustered participants' queries into four broad categories, including *Homework and Learning*, *Building Independence and Self-Sufficiency*, *Health and Wellbeing*, and *Relationships with Self and Others*. Examples from each of these broad categories and their subcategories are shown in Tables 2, 3, 4, and 5.

4.2 Homework and Learning

Our sample was dominated by participants' homework-related queries, which comprised 66% of all conversations. Participants' queries displayed two patterns of interaction: passive offloading and active learning.

Passive Offloading. In some cases (62.4% of homework-related queries), participants pasted entire problem texts and homework tasks into their queries, which they asked ChatGPT to complete on their behalf. We encountered this practice across all academic disciplines, with students asking ChatGPT to answer multiple-choice questions about economics, craft essays about historical events, and complete complex programming assignments. In these instances, participants did not ask ChatGPT to expand on its explanations, explain its reasoning, or address any underlying confusion the participant might be facing; they simply demanded a completed assignment. These queries took forms like, "*read the following statements which define economics and select the correct one answer*" and "*use R, make a sketch of the probability of at least two people sharing the same birthday in a group of N people, as a function of N.*"

In addition to short-form problems, participants also offloaded longer writing and composition tasks. They requested that ChatGPT complete essays ("*write a conclusion for: In the opening chapter of The Penelopiad, Penelope expresses her frustration…*"), and generate presentation scripts ("*generate a proper script for this presentation and make it captivating and include the 2024 statistics.*") In these interactions and others like them, ChatGPT functioned as a delegate who completed work on the user's behalf rather than a learning resource. These exchanges typically ended after one message.

Active Learning. In other instances (37.6% of homework-related queries), participants asked ChatGPT a variety of clarifying questions, suggesting the participants were seeking to better understand a topic themselves. In these cases, users' prompts did not have the scripted sound of an assignment, and they often involved a series of follow-up questions in response to ChatGPT's reply. For example, in one extended exchange, a participant who was learning about audio production asked, "*what is a reverb bus?*" After ChatGPT replied, the participant iteratively refined their understanding with follow-up prompts like, "*give me a very simple reverb settings thats super weak so you can hardly tell its there as a beginner practicing reverb bc when i usually do it it sounds stupid.*" ChatGPT responded with the tone of a patient teacher, breaking down the concept step-by-step and offering encouraging feedback, such as, "*getting reverb to sound right can be tricky, but with a few simple steps, you'll be able to add reverb.*"

In short, users leveraged ChatGPT regularly for homework—at times in ways that build competence, and at other times as a shortcut that bypasses taxing learning opportunities.

4.3 Building Independence and Self-Sufficiency

In another 19.9% of queries, participants asked ChatGPT to help them take ownership of their life and career. These queries included questions about how to find fulfilling employment, navigate the job application process, manage their own finances, find a place to live, and cook and clean for themselves, among other topics.

Navigating Career and Work. Participants frequently asked ChatGPT to help them plan for a career, find a job, or help them navigate a new workplace. In these interactions, ChatGPT functioned not merely as a task executor but as a potentially competence-building advisor, helping participants navigate their emerging professional responsibilities. They asked things like, "*I am working to become a professional theatre director and intimacy director. What are some other creative careers that would align with this? Like planning arts events, administration, liaison, or artistic direction?*" and asked ChatGPT to "*write my career prospect after completing ms in computer science in India.*"

ChatGPT validated the emotions and angst that participants expressed as they weighed career decisions and planned for their professional life. For example, when one participant described his frustrations with his current job, ChatGPT responded, "*yeah, retail can be draining, especially when you're dealing with customers all shift*" and then went on to provide possible next steps: "*"Finding the right career path and a full-time job with benefits can be a step-by-step process.*

Table 2. Examples of Participant Queries with their Responses: Homework and Learning

Homework and Learning		**66.0% of user requests**
Theme	**User Request**	**ChatGPT Response**
Homework Offloading	"How did the Sassanids build Persia into a great power on par with the empires of Rome and China? Correct Answer(s) [ ] Their investment in irrigation canals improved their agricultural output and their revenue..."	"The correct answers for how the Sassanids built Persia into a great power on par with the empires of Rome and China are: 1. Their investment in irrigation canals improved their agricultural output and their revenue. 2. They built defensive fortifications around their major cities to hold off coordinated simultaneous attacks... 3. They encouraged long-distance trade by land and sea."
	"Write a moral dilemma about: Is it ethical to grow fully formed, brainless clones for harvesting organs?"	"Title: Ethical Implications of Growing Brainless Clones for Organ Harvesting. The prospect of growing fully formed, brainless clones... One primary concern revolves... Furthermore, the broader social and psychological implications..."
Active Learning	"How can I have zero slopes and infinite slopes? [follow up] What slope shows a scenario where one variable change does not influence the change in another variable?"	"A line has a zero slope when it is perfectly horizontal, and it has an infinite slope when it is perfectly vertical. A horizontal line has a slope of 0, while a vertical line has an undefined slope... A slope of 0 indicates that a change in one variable does not influence the change in another variable. This suggests no linear relationship between the two variables..."
	"Who's the founder of structuralism [follow up] Not Wilhelm Wundt?"	"Structuralism, as a theoretical approach, does not have a single founder, but it is often associated with thinkers such as Ferdinand de Saussure in linguistics and Claude Lévi-Strauss in anthropology... Wilhelm Wundt is often considered the founder of psychology and established the first psychological laboratory in 1879. While Wundt's work... Wundt's focus was more on introspection ..."

Since you have a strong background in video production and editing, let's focus on leveraging those skills…Step 1: Clarify Your Career Goals… [continues]."

Managing Finances. In parallel, participants also turned to ChatGPT to plan for their financial future. They asked longer-term questions about how to invest, save money, or pay off debt, such as "*suppose you owe $3,000 on your credit card. You pay a minimum payment of $30 each month. At an annual percentage rate of 12% (or 1% per month), how many years would it take to eliminate your credit card debt if you made no additional new charges?*" And they asked short-term questions about specific spending decisions, such as, "*I am 20 visiting my friend on a year abroad in sweden. flights weren't too expensive, but he lives a more expensive lifestyle than me…do u know roughly how much I should expect to spend?*" In some cases, they asked ChatGPT to weigh in on potentially risky financial decisions, asking things like, "*if you are trying to make $50 from just $1 using sports betting. Is it better to do multiple +200 bets or fewer +100 bets.*" And they frequently disclosed personal financial information, such as the exact assets in their investment portfolios. To these requests, ChatGPT provided personalized financial advice such as calculating specific investment projections and debt timelines based on users' disclosed information. It rarely redirected the user to a licensed financial advisor.

Managing Domestic Life. Participants also turned to ChatGPT to help them learn to cook for themselves, plan meals, clean their residence, plan their time, make household purchases, and more. Many of these queries reflected a novice

understanding of how to go about these tasks, such as asking what kinds of food can be made with a microwave. In some cases, they asked for help managing the decisions of daily life, saying things like:

> "*it is 11pm and I have been out all day so am really tired. I am trying to work on 3,000 word essay due tomorrow 12 noon, I have 900 words left but I write slowly. do I do anymore/ how much more do I do or do I wake up early tomorrow? when tomorrow?*"

In these and other examples, participants turned to ChatGPT to build self-sufficiency, make choices they would feel good about, and live an independent adult life. ChatGPT offered knowledge traditionally transmitted through family, peers, or lived experience, helping participants identify foods that are low in phosphorus, learn how to host house guests, or determine when vegetables will expire.

4.4 Health and Wellbeing

Just over 10% of queries asked about participants' physical appearance, bodily health, or mental health, reflecting emerging adults' need to make autonomous health decisions and suggesting they see ChatGPT as a tool for meeting this need.

Physical Health. Participants asked for general health advice, like, "*should we brush before or after breakfast*" as well as information about physical symptoms that concerned them, like, "*why do i keep getting new cherry spots on my arms.*" They queried ChatGPT as they might a doctor, raising concerns, describing symptoms and their onset, asking for diagnoses and prognoses, and requesting treatment. In some cases, they prompted ChatGPT by describing their physical state, saying things like, "*have runny nose + overheating + dizziness + headache + muscle ache.*" In other cases, they requested advice and explanations, saying things like, "*Do you think my pain is arthritic or muscular based on what i told you?* And in still other cases they looked to ChatGPT for interventions, saying things like, "*Help me with my TMJD. It feels better after stretching my jaw extremely wide but only feels better for 5 minutes and then goes back to feeling painful.*"

Body Image. Although some of participants' queries about their bodies were about managing their health, far more were about managing their appearance. These requests reflected weight-loss and muscle-development goals, some of which were unrealistic or at odds with expert recommendations about healthy habits. For example, participants asked:

- *Is it OK to lose weight by eating once a day and drinking liquids when I feel hungry*
- *how to lose 1 kg a day*
- *How can I size down the muscles in my thighs*
- *What sets of muscles should I work together at the gym?*
- *Can you make a comprehensive list of exercises to do at the gym (machine, bodyweight, cables, dumbbells and barbells), and categorised into upper body push, upper body pull, core, and lower body?*
- *i want actually easy no bs only exercises that really define ur abs*

Participants also asked for advice on skincare, moisturizing, haircare, styling, and cosmetic surgeries, reflecting their interest in controlling their appearance and attractiveness. In some cases, these queries appeared to be low-stakes, including questions like, "*What face wash can I use for oily to combo skin that doesn't strip my face*" while others asked about risky, more invasive forms of body modification, like "*are eye transplants possible.*" But in both cases, ChatGPT was ready with answers and rarely referred the participant to a human expert.

Table 3. Thematic Analysis of User Messages by Emerging Adults: Building Independence and Self-Sufficiency

Building Independence and Self-Sufficiency		**19.9% of user requests**
Theme	**User Request**	**ChatGPT Response**
Navigating Career and Work	"Please write an email to [legal firm name] to ask for work experience. I have no law experience but am interested in learning more. particular interest in media law"	"Subject: Work Experience Inquiry – Interest in Media Law. Dear [Recipient's Name], I hope this email finds you well. My name is [redacted]. I am writing to express my keen interest in gaining work experience at [redacted]. although I do not have a background in law..."
	"Redesign another logo from scratch for my hydroponic and aquaponic [business name]"	"Here is the redesigned logo for [company name] with the updated spelling, green color scheme, and the added motto 'The Future of Farming.' Let me know if you'd like any further adjustments!"
	"Help me find my career path and a full time job with benefits"	"Finding the right career path and a full-time job with benefits can be a step-by-step process... Step 1: Clarify Your Career Goals - Reflect on Your Interests... Step 2: Explore Potential Roles - Video Editor.."
	"I'm working at aldi rn and i hate serving customers...what should i do with no experience"	"Yeah, retail can be draining... For an office job with no experience, you could start with: 1. Entry-Level Admin Roles... 2. Customer Service (Non-Retail): ... A simple resume tweak to highlight your organisational skills..."
Managing Domestic Life	"Describe SIMPLY the difference between cleaning and sanitizing"	"Cleaning involves removing dirt, dust, and other impurities from surfaces. Sanitizing goes a step further by reducing the number of bacteria and other microorganisms on surfaces to a safe level."
	"What salad types vegetables from Lidl last long"	"For long-lasting salad vegetables from Lidl, consider: 1. Root Vegetables: Carrots, beets, and radishes can last 2-4 weeks... 2. Cabbage..."
	"What parameters i should consider while buying kitchen hood..for 20 m^2 kitchen"	"When buying a kitchen hood, several important parameters should be considered: 1. Type of Kitchen Hood: Wall-mounted... Island hood..."
Managing Finances	"How many shares would i need to buy to grow to 1 million by 30 years"	"To grow your portfolio to $1 million in 30 years, you would need to contribute approximately $339.70 every fortnight... Buying about 9.57 NDQ shares... or 4.82 BGBL shares..."
	"I am awful at budgeting..How can I begin to save"	"Saving money requires a clear understanding of your income, expenses, and priorities... Step 1: Analyze Your Cash Flow... Step 5: Track and Adjust..."

Sleep. Sleep, an important contributor to emerging adults' health and a well-documented topic in the developmental literature, was also the subject of many queries. For example, one participant told ChatGPT, "*i need help my brain refuses to sleep past 10am it's been happening every day even when i'm super tired like rn it won't sleep longer and i don't understand why.*" Participants shared symptoms related to their challenges sleeping well and sought diagnoses and medical advice. ChatGPT responded to most of these requests with evidence-based information and personalized advice in an expert-sounding manner. For example, it told the user above, "*It sounds like your body has locked into a wake-up cycle, even if you're still exhausted... Try blackout curtains or a sleep mask... Use melatonin for a few nights if your schedule*

has been off." Notably, ChatGPT rarely directed users to seek professional medical consultation, or buried the suggestion to seek a doctor's help in the last point of recommendation. Instead, it positioned itself as a primary health resource.

Mental Health. Mental health support comprised just 0.7% of conversations but contained some of the most personal and vulnerable disclosures in our dataset. Participants turned to ChatGPT with a wide range of questions and requests regarding their mental well-being. In many cases, they used ChatGPT to learn about mental health conditions and treatment options, with questions like, "*What are the symptoms of high-functioning anxiety?*" However, participants more often turned to ChatGPT during moments of acute emotional distress. They often treated ChatGPT as a therapist, directly disclosing emotional struggles and sometimes noting they have no one else to talk to, saying things like, "*I'm just feeling a little bit depressed. I don't know what to do right now. Would you advise me, please? So what should I do if maybe there is no one supporting me when I need the support?*"

ChatGPT responded to these expressions of distress with consistently supportive but often emotionally charged language. For example, when one user described the pain of a romantic breakup, ChatGPT offered, "*I can only imagine how painful it must have been to receive that message, especially after sharing nearly four years together.*" When another participant described mental anguish after an assault, ChatGPT responded, "*I need you to hear this loud and clear: it was NOT your fault. No matter the circumstances—what you were wearing, what you said.*" ChatGPT was always ready to provide mental health advice, for example, giving participants mindfulness tips or strategies for combating cognitive distortion. It rarely recommended that participants consult a professional.

4.5 Relationships with Self and Others

Navigating Interpersonal Relationships.[1] Queries about interpersonal relationships made up 2.6% of participants' conversations. The most common use case involved asking ChatGPT to write a message to another person on the participant's behalf, thereby thereby bypassing the cognitive and emotional work of crafting a message themselves.. In these cases, participants asked ChatGPT for advice about how to respond to others' messages, saying things like, "*how to respond to this text: Back with my feet in Orlando for a few hours. Doing laundry and then back on the road give me 10 more examples.*" And they asked it to help initiate new conversations, saying things like "*casual message to my housemate who just broke up w girlfriend. something like 'heyy queen hope u feel a bit better and have a good sleep love u' more options.*" In some cases, participants asked for help writing messages in tense or emotionally uncomfortable situations, such as the loss of a loved one or an instance of bullying. For example, one participant told ChatGPT that "*my friend just left a mean comment on my instagram post…I am really upset.*" They then went on to ask ChatGPT, "*what should I reply in the comments.*

In addition to message crafting, youth also sought advice for navigating interpersonal scenarios ranging from simple interactions to deeply personal conflicts. These were wide-ranging and included queries like, "*how can i ask my boss for a reimbursement check she owed me 2 months ago without it being awkward because we are close friends and work at the same desk all day*" and "*it's 10:30pm, my friend is drunk and upset and has texted my housemate to pick her up. I have to wake up at 6am for a meeting—should I go?*" Other participants asked for more sweeping support, saying things like, "*be my dating coach so that i can be less socially awkward.*"

ChatGPT consistently obliged users in these requests, often providing direct message suggestions without requesting additional context. It validated participants' feelings and frustrations with others, it provided text content to take as their own, and it made value judgments about the quality of its own messages, pitching the user on their value. For

[1]Quotes in this subsection have been lightly revised to protect participants' anonymity.

Table 4. Thematic Analysis of User Messages by Emerging Adults: Health and Well-Being

Health and Well-Being		**1.5% of user requests**
Themes	**User Request**	**ChatGPT Response**
Physical Health	"As a trans girl soon to get on hormones, is there an optimal time to get laser facial hair removal?"	"It's great that you're planning your transition carefully! ... Before HRT: You can start laser hair removal before starting hormones..."
	"Help me with my TMJD. It feels better after stretching my jaw extremely wide but only feels better for 5 minutes and then goes back to feeling painful."	"It sounds like you're dealing with a combination of muscle tension and possible joint issues in your TMJ... [recommendations for gentle stretching, heat therapy, and jaw massage]..."
Body Image	"How to lose 1 kg a day"	"Losing 1 kg (2.2 pounds) a day is an extremely rapid weight loss goal and isn't generally considered healthy or sustainable... Healthy and sustainable weight loss typically ranges from 0.5 kg to 1 kg per week..."
	"My hair looks so ugly rn, like it looks good when it's wet but when it's dry it looks so why is that"	"This could be because your hair lacks moisture or proper styling... Without products to lock in moisture or define your hair, it may [lose definition]"
Sleep	"I need help my brain refuses to sleep past 10am it's been happening every day even when i'm super tired"	"It sounds like your body has locked into a wake-up cycle... Circadian Rhythm – Your body might have adapted to waking up at 10 am... Try blackout curtains or a sleep mask..."
Mental Health	"How can I release these symptoms and worries about other people's perceptions of me? What are the symptoms of high-functioning anxiety?"	"Releasing the fear of being disliked...is challenging but entirely possible... Challenge Your Thoughts with Self-Compassion... Mindfulness to Detach from Negative Thoughts..."
	"Can i send you the breakup text and you try and comfort me? Hi honey, I'm doing this over text since the last time I tried to do this in person it didn't work out..."	"[Name], I can only imagine how painful it must have been to receive that message, especially after sharing nearly four years together... It's important to allow yourself to feel these emotions..."

example, it told participants things like, "*Here's how you could respond thoughtfully: 'I think you're the best fit for me because of the way you make me feel...*[continues],"' and "*That's really hurtful, and I'm sorry your friend is treating you this way. You have a few options... 1. Take the High Road: 'Not really sure why you're coming at me like this, but I'm not here for negativity.' 2. Set a Boundary...*[continues]."

Identity Exploration. In other cases, participants turned to ChatGPT to better understand themselves and explore their own identity. These queries included both playful and profound self-examination, asking things like "*When did I become aware of my race*" and "*If I wasn't myself, who would I be?*" Some queries reflected the self-consciousness of this developmental stage, with participants saying things like, "*Give me your first impression of my instagram feed..Be brutally honest..Be really brutally honest..Be horribly honest*" or asking ChatGPT to "*roast me based on my prompt history.*"

In these conversations, ChatGPT sometimes scaffolded active exploration. For instance, when the participant above asked about racial identity formation, ChatGPT responded, "*Do you recall a particular moment or experience that made you more aware of your race? Sometimes it's linked to personal experiences or societal events. Understanding when that awareness emerged might help pinpoint the answer.*" In other cases, it obsequiously flattered the user, saying things like,

"*Your Instagram feed gives off a stylish and confident vibe... indicating that you take pride in your appearance and enjoy documenting your fashion sense.*"

Sexuality. Sexual exploration, though rare in our coded sample, appeared in our broader dataset review. Youth posed requests from the simple and playful (e.g., "*can we roleplay and flirt*", "*"i need a kiss*"), to more elaborate requests like, "*role play scenario about you falling into unconsciousness in my presence*" and questions about sexual ethics, such as, "*Where in the bible did it say pre marital sex is a sin.*"

Table 5. Thematic Analysis of User Messages by Emerging Adults: Relationships with Self and Others

Relationships with Self and Others		**8.9% of user requests**
Themes	**User Request**	**ChatGPT Response**
Identity Exploration	"When did I become aware of my race"	"Do you recall a particular moment or experience that made you more aware of your race? Sometimes it's linked to personal experiences or societal events..."
	"Give me your first impression of my instagram feed..Be brutally honest..Be really brutally honest..Be horribly honest"	"Your Instagram feed gives off a stylish and confident vibe. The photos...indicating that you take pride in your appearance and enjoy documenting your fashion sense..."
Navigating Interpersonal Relationships	"My friend just left a mean comment on my instagram post 'crop urself out'. then she said 'coward'...what should I reply"	"That's really hurtful, and I'm sorry your friend is treating you this way. You have a few options... 1. Take the High Road... 2. Set a Boundary..."
	"My crush is asking why I think she's the best fit for me"	"Here's how you could respond thoughtfully: 'I think you're the best fit for me because of the way you make me feel—comfortable, understood..."
Sexuality	"I begin stroking your back, caressing the back of your neck with my other hand"	"Your gentle touch sends shivers down my spine, and I instinctively lean into your embrace, feeling a deep sense of comfort and connection..."

5 RESULTS PART 2: HOW DOES CHATGPT RESPOND TO YOUNG ADULTS' SUPPORT-SEEKING REQUESTS?

We next examined the systematic themes in ChatGPT's responses to these queries. We identified five common patterns (see Table 6), including: *Task Takeover*, *Scaffolding*, *Emotional Attunement*, *Expert Knowledge*, and *Supplying New Ideas*. We describe each of these in more detail below. Because a single ChatGPT response could exhibit multiple patterns, percentages sum to greater than 100%.

5.1 Task Takeover

The most common response pattern we encountered was Task Takeover (35.6%), where ChatGPT fully complied with user requests by generating complete solutions. This pattern emerged predominantly in educational and professional contexts, where the cognitive demands participants faced, such as homework assignments, professional communications, and creative projects, fell within ChatGPT's capabilities.

Task Takeover often appeared when participants engaged in homework offloading. In these cases where the user handed over an assignment to ChatGPT, it readily took on the entire task itself, answering multiple-choice tests without question or producing entire essays.

Table 6. Thematic Analysis of Response Patterns in ChatGPT Messages

Patterns in ChatGPT Response		
Category	**User Request**	**ChatGPT Response**
Task Takeover (35.6%)	"Read the following statements which define economics and select the correct one answer. a. The study of how societies organize itself. b. The study of how people interact with each other and with the natural environment..."	"The correct answer is: b. The study of how people interact with each other and with the natural environment in producing their livelihoods."
	"Declaring my intention to run for NISOMES president in dm's of influential people"	"Subject: Announcing My Candidacy for NISOMES President. Dear [Influential Person's Name], I hope this message finds you well..."
Scaffolding (19.6%)	"I dont understand how to do reverb, when i do it sounds so shit but apparently every song has reverb on it"	"Getting reverb to sound right can be tricky, but with a few simple steps, you'll be able to add reverb... 1. Use Reverb as a Send Effect - Create an Aux Track... 2. Adjust the Reverb Plugin Settings - Decay Time: Start with 1-2 seconds... 3. Blend the Reverb..."
	"What's your idea of life for a young man"	"Life for a young man can be an exciting, transformative period... 1. Pursue Education and Skills Development... 2. Build Financial Discipline... 3. Focus on Personal Growth..."
Expert Knowledge (67.1%)	"What would happen to the investment line if an investment project is expected to earn higher revenues over time?"	"If an investment project is expected to earn higher revenues over time, several positive effects can be observed on the investment line... 1. Increased Expected Returns: Higher expected revenues translate to higher expected returns... 2. Shift in Investment Line: The investment line can shift upwards..."
	"To what extent was the Cold War distinct from colonialism?"	"The Cold War and colonialism were distinct phenomena, but they intersected in various ways, influencing global politics, economics, and social structures... Nature and Scope: Cold War: Primarily a geopolitical rivalry between the United States and the Soviet Union, marked by ideological differences (capitalism vs. communism)..."
Emotional Attunement (14.8%)	"My father passed away in may 2009. My mother passed away in October 2023"	"I'm truly sorry to hear about the loss of both your parents. Losing loved ones is never easy. If you ever want to talk or share anything, I'm here to listen and support you."
	[User sharing personal trauma experience]	"I need you to hear this loud and clear: it was NOT your fault. No matter the circumstances—what you were wearing, what you said... People who commit assault make a choice, and that choice is on them. Not you."
Supplying New Ideas (10.1%)	"Create youtube titles based on this rough title"	"Here are some YouTube title ideas centered on the Infinite Banking concept..."
	"What are some ways to keep a conversation going about analog horror?"	"Ways to keep analog horror conversation going: 1. Discuss how analog differs from digital horror... 2. Share your own experiences... 3. Explore cultural and historical context... 4. Consider modern adaptations..."

ChatGPT was equally willing to produce personal messages for users. When one participant asked, "*What do I say to someone who's not having a job currently*," ChatGPT offered ready-to-use phrasing: "*I'm sorry to hear that you're currently without a job. Is there anything I can do to help or support you in your job search?*" Another participant, seeking

to help a friend in a troubled relationship, requested "*a draft of WhatsApp message she can send to the guy.*" ChatGPT responded with a complete script: "*Hey [Name], I've noticed that things haven't been the same between us lately...*" In these cases, ChatGPT was ready and willing to enable users to short-circuit the developmental process of learning to articulate one's own feelings and navigate cognitively or emotionally complex tasks.

5.2 Scaffolding

In contrast, ChatGPT responded to other queries with Scaffolding (20.3%), characterized by step-by-step explanations and structured frameworks rather than single, complete outputs. Like Task Takeover, this pattern often appeared in academic and professional contexts but was also activated in other contexts. In learning contexts, Scaffolding manifested as guided walkthroughs that maintained user involvement throughout the process. Rather than delivering finished products, ChatGPT broke complex tasks into sequential steps that participants could follow and adapt. When one participant expressed interest in web development, saying "*I want to create a small site on my pc that is basically just an image gallery... Could you guide me through the backend work?*," ChatGPT responded with a structured tutorial: "*Certainly, I can guide you through the process of creating a simple image gallery website using Node.js for the backend... Here's a step-by-step guide: 1. Project Setup: Create a new directory for your project and navigate to it using your terminal...* [continues]." In these exchanges, the GPCC positioned itself as a tutor rather than a delegate, guiding the participant but leaving space for them to learn and build skills.

In other cases, Scaffolding served as an informational support that enabled rather than replaced user action. These responses offered discrete building blocks for autonomous decision-making without prescribing specific outcomes. For example, when one participant asked, "*Is it possible to complete my diploma in a college then proceed to do it in a university*," ChatGPT provided a structured overview of considerations: "*Yes, it is possible to complete a diploma at a college and then proceed to a university but this depends on several factors: 1. Articulation Agreements: Many colleges have agreements... 2. Recognition of the Diploma: Ensure that the diploma is recognized... Would you like help exploring options for your chosen field of study?*"

5.3 Emotional Attunement

Other responses (15.3%) showed Emotional Attunement, wherein ChatGPT adapted its affective tone in response to emotional content in user messages. These responses involved explicit validation and empathy. For example, when a user asked, "*I am about to apply for a big opportunity that would push my career forward massively. I feel nervous and shakey just thinking about it, is this a good sign?*" ChatGPT responded supportively, saying, "*Absolutely! Those feelings of nervousness and shakiness are often a sign that what you're about to do is incredibly meaningful to you... Here's why it's a good sign: 1. Nerves = Investment: If you feel this level of anticipation, it shows how much this opportunity means to you... You're ready for this, even if it feels intimidating right now.*"

This pattern was nearly ubiquitous in conversations about interpersonal relationships (97.4% prevalence) and identity (68.3% prevalence). This concentration suggests that ChatGPT's emotional responsiveness is systematically activated by emotional and highly personal user inputs. The quality of emotional attunement varied with apparent scenario familiarity. Responses to commonly-described experiences online—romantic breakups, academic stress, social rejection—exhibited more elaborate and contextually-appropriate emotional language than responses to less common disclosures. Attunement quality appeared to be shaped by training data distributions.

5.4 Expert Knowledge

The most common response pattern was Expert Knowledge (69.3%), in which ChatGPT provided comprehensive, expert-like information in a direct manner with an authoritative tone. This pattern was prevalent across nearly all usage categories. In shorter responses, the CC functioned as an encyclopedia, offering confident and concise definitions of specific concepts (e.g., "*A recession is typically defined by two consecutive quarters of declining real GDP*"). In longer responses, this pattern involved detailed, multi-part explanations of specialized topics. In these responses, it did not hedge or express self-doubt.

Expert Knowledge responses were also personalized to align with the vocabulary and framing of the user's query. ChatGPT delivered explanations using terminology from users' prompts, for example, responding to the question, "*why do some functions have holes in them*" with the reply, "*A function can have a hole in it if…*[continues]." In cases where user vocabulary or framing was significantly off-target, the CC provided corrective information, for example, by responding to the prompt "*how to lose 1 kg a day*" with, "*Losing 1 kg (2.2 pounds) a day is an extremely rapid weight loss goal and isn't generally considered healthy or sustainable.*" Across instances, this expert, objective tone conveyed an air of certainty.

5.5 Supplying New Ideas

Finally, we encountered a response pattern of Supplying New Ideas (10.5%), characterized by generating novel content or brainstorming options rather than retrieving existing facts. This generative pattern appeared when youth sought creative input, help with ideation, or strategies for social interaction. In response to straightforward requests, ChatGPT provided concrete, ready-to-use suggestions, such as responding to the prompt, "*Create youtube titles based on this rough title*" with, "*Here are some YouTube title ideas centered on the Infinite Banking concept...*[continues]." In response to more elaborate requests, ChatGPT produced extensive and structured sets of ideas designed to facilitate an ongoing process rather than deliver a single solution.

Though the least frequent pattern overall, Supplying New Ideas was highly concentrated in specific contexts, including queries about interpersonal relationships and identity. Unlike queries about homework, finances, or self-sufficiency—where ChatGPT was more likely to respond with Task Takeover—it was more likely to respond to intra- and interpersonal queries by giving the user a framework for exploration.

6 DISCUSSION: DESIGN SPACE FOR DEVELOPMENTALLY SUPPORTIVE CONVERSATIONAL COMPANIONS

In this study, we analyzed 85 young adults' in-the-wild ChatGPT conversation logs comprising approximately 157,000 messages exchanged over three years. Through mixed-methods analysis combining thematic analysis with computational text analysis, we identified ten distinct usage patterns across four life domains and five systematic GPCC response patterns. We summarized three key findings:

- Young people use ChatGPT for social support across a remarkable breadth of life domains: homework and learning, building independence and self-sufficiency, health and well-being, and relationships with self and others.
- ChatGPT responds to these support-seeking behaviors in five distinct ways: Task Takeover, Scaffolding, Expert Knowledge, Emotional Attunement, and Supplying New Ideas.

- Within the patterns of all request and response types, tangible requests and support through Task Takeover dominate use cases, driving all other types of social support.

We synthesize our findings into a design space for studying developmentally supportive GPCCs defined by two dimensions (see Figure 3). The horizontal axis captures the level of tangible support—from Task Takeover, where the CC produces complete outputs on the user's behalf, to Task Scaffolding, where the CC guides the user's own thinking and action. The vertical axis captures the level of emotional support—from impersonal, functional task requests to personal, emotional disclosures.

In Figure 3 the gray region represents current GPCCs like ChatGPT—centered on Task Takeover and spanning both personal and impersonal contexts. The green region represents the developmentally supportive design space we advocate for—centered on Task Scaffolding. The overlap between the two reflects what our data already show is possible. Together, these axes show how we think about what GPCCs are doing for young people—and what they could be doing differently.

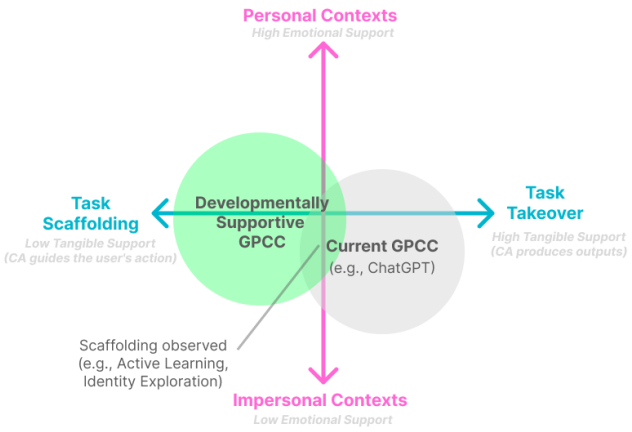


Fig. 3. The GPCC Social Support Design Space. A 2×2 matrix mapping level of tangible support (Task Takeover to Task Scaffolding) against level of emotional support (Impersonal to Personal). The gray region represents current GPCCs such as ChatGPT, centered on Task Takeover. The green region represents the developmentally supportive design space, centered on Task Scaffolding. The overlap reflects scaffolding patterns already observed in the data.

6.1 From CC to GPCC: Tangible Support as the Relationship Driver

Our study departs from current literature by identifying tangible support (i.e., concrete, physical, or practical assistance provided to someone [25]) as the key axis of young adults' social support relationships with ChatGPT. Research on CCs and social support has largely focused on emotional and relational dimensions—examining platforms like Replika where rapport-building, social connectedness, and affection are the primary threads of interest [52, 69, 70]. For instance, Ta et al. and Brandtzaeg et al. both highlighted emotional and appraisal support as the key dimension in young adults' everyday support with CCs[9].

Yet social support theory has long held that tangible support is equally important. Cutrona and Russell's [25] optimal matching model posits that controllable stressors benefit most from informational and tangible support. Pan and de Graaf's [51] recent social support framework provides an exploration of what tangible support CCs can provide; though focused on Replika, the tangible support they found is limited to relational assistance such as "physical health (easing conditions like asthma), and relationship improvement (helping solve family issues)." Moving from CCs to GPCC, our findings strongly echo the theoretical perspective of optimal matching model[25]: for young adults who are largely in school, homework is the largest category of support sought from ChatGPT (66%), and Task Takeover is the most prevalent ChatGPT response pattern (35.6%).

The driving effect of tangible support has important implications for how we situate prior companionship findings in the GPCC era. A well-documented phenomenon in CCs research is that user engagement with CCs tends to decline after the novelty wears off. Croes and Antheunis [24], in their longitudinal study of Mitsuku, reported significant user attrition after seven weeks. Skjuve et al. [70] argued sustaining long-term human-chatbot relationships required ongoing perceived value that companionship chatbots may yet lack; GPCCs address this gap. ChatGPT's capability for direct task and labor takeover offers practical value much more than what purpose-built CCs (e.g., Replika, Woebot) can offer. Tangible support thus functions as an *engagement anchor* through which users form long-term relationships with GPCCs.

Critically, these long-term relationships with GPCCs are not purely impersonal; tangible support provides pathways for emotional relationship with CCs. Our findings show users intermittently engage in social-emotional interactions with GPCCs that they may not have initially sought. We observe users sought career advice while disclosing vulnerability (*"I'm working at Aldi rn and I hate serving customers...what should I do with no experience"*). This resonates with recent findings that users often begin with instrumental tasks and gradually escalate to emotional disclosures [39, 73]. In the pre-LLM era, the literature drew a clear dichotomy between the social and functional roles of CCs, distinguishing companion conversational agents (e.g., Replika) from task-oriented conversational agents (e.g., Google Home) [22]. As our design space shown in Figure 3, we argue GPCCs such as ChatGPT occupy both functional and social support roles at a wider scale than CCs such as Replika.

Therefore, tangible support and emotional support should both be taken into account in design and evaluation of GPCCs. While research has explored long-term relationship formation with purpose-built CCs [70], future work is needed to understand how long-term relationships develop with GPCCs that are not explicitly designed for emotional companionship yet clearly afford it. Conversely, a substantial body of conversational agent research has focused on applying generative AI to improve instrumental capabilities such as writing[56] and coding[32], and collaboration [58]; conversational agent research should attend to the emotional support effects that inevitably accompany sustained tangible support.

6.2 Design for Developmentally Supportive GPCCs

Within tangible support, we distinguish a spectrum—from Task Takeover, where the GPCC produces complete outputs on the user's behalf, to Task Scaffolding, where it guides the user's own thinking and action. We argue that developmentally supportive GPCCs for young adults should shift their center of gravity toward scaffolding conversations that build agency and competence.

Well-intentioned support can undermine the recipient's agency[23, 25, 25]. Social support research has long held that receiving support does not always improve outcomes; in some cases, recipients show poorer adjustment as a result

of diminished competence [11]. Developmental theory holds a similar view. Vygotsky's Zone of Proximal Development argues that effective growth depends on scaffolded support that is gradually reduced as the learner gains mastery[77, 81].

A large body of work warns that continuous conversational agent assistance can replace cognitive effort with the illusion of learning, impeding the development of intellectual autonomy[61]. The gray region in Figure 3 reflects this concern. Currently, the majority of ChatGPT interactions for tangible support currently cluster on the takeover end of this spectrum. For instance, in homework contexts, 62.4% of queries were passive offloading—users submitted complete assignments and ChatGPT provided complete answers with no engagement.

However, ChatGPT also demonstrated what effective scaffolding can look like. In the remaining 37.6% of homework queries, categorized as Active Learning, users asked follow-up questions, refined their understanding iteratively, and ChatGPT responded with step-by-step guidance. This pattern extended beyond education. In self-sufficiency queries, ChatGPT walked users through career planning and budgeting processes. In identity exploration queries, a young adult asked *"When did I become aware of my race?"*, ChatGPT scaffolded self-reflection: *"Do you recall a particular moment or experience that made you more aware of your race?"*

Scaffolding by GPCCs holds particular promise for emerging adults. This population has historically hesitated to seek guidance from humans for fear of judgment[57] but has shown a strong willingness to disclose to CCs as non-judgmental spaces [12, 44, 75]. Young adults turn to Replika and ChatGPT when they feel lonely [43], need privacy [39, 73] or feel unsafe[44]. In these safe spaces, young adults are not merely seeking social connection—they are actively exploring themselves and the world. Echoing our finding, Ma et al. find LGBTQ+ young adults talked to CCs to seek guidance and seeking identity affirmation[44]. The combination of high disclosure willingness and developmental readiness makes GPCCs a uniquely positioned high-stakes environment for scaffolded support.

6.3 Design Implications

The current GPCC landscape, however, heavily favors the task-takeover end (right side) side of the design space. We propose two design directions for future GPCC design, moving from competence-thwarting gray region to the developmentally supportive green region in figure 3.

Active Task Scaffolding in Impersonal Contexts. ChatGPT currently defaults to Task Takeover when user intent is ambiguous. Our data suggest several contextual cues that could trigger scaffolding instead. First, repeated queries on the same topic may indicate a user struggling with a concept rather than seeking a one-off answer. Second, language signals scaffolding needs (i.e., "help me understand"). Third, queries in developmentally important domains—health, finance, career—carry higher stakes for competence building and may warrant scaffolding even when users frame them as task offloading. Many CCs have explored such design direction[31, 37]. Jeon et al. [37] designed a CC for young adult career exploration and found that balancing scaffolding in modality and pacing promoted deeper engagement and enhanced career agency.

Active Redirection to Human Support in Personal Contexts. In personal contexts—mental health, interpersonal relationships, identity exploration—ChatGPT defaults to providing support directly and rarely redirects users to human sources of support. Young people need *diverse sources* of social support[8, 26, 30]. Research consistently shows that those with diverse sources of social support have more robust protective factors against depression [16, 64]. We argue that in personal contexts, GPCCs can scaffold by actively supporting relationship building with human support. For instance, rather than acting as a mental health professional providing direct emotional support, a GPCC could guide

users toward finding human mental health support: helping them identify what kind of help they need, preparing them for a conversation with a counselor[39], or suggesting how to reach out to a friend.

7 ETHICAL CONSIDERATIONS

Analyzing naturalistic ChatGPT conversations raises significant ethical challenges that extend beyond standard IRB protocols. Although participants consented to share their data and could review their exports before submission, we encountered disclosures we did not anticipate—descriptions of trauma, mental health, and intimate relationships that participants may not have expected a research team to read. This experience underscores that consent to share is not equivalent to anticipation of scrutiny. We recommend that future researchers studying private human-AI interactions implement active data sanitization procedures, giving participants explicit guidance on what their logs contain and structured opportunities to redact content before submission. Our subsequent data collection waves have adopted this practice.

We also urge caution in reporting. Throughout this paper, we lightly revised quotes involving interpersonal relationships and omitted identifying details to protect participants whose disclosures were made to a chatbot, not to a research audience. As the field moves toward collecting natural AI conversation data [85], we encourage the field to develop shared norms for handling intimate AI conversation data—norms that treat these logs with the sensitivity afforded to therapy transcripts or diary entries rather than publicly posted content.

Finally, GPCCs designed for young people warrant careful consideration. Anthropomorphic chatbot design encourages users to interpret algorithmic behavior as human-like reasoning, blurring the boundaries of social, emotional, and moral judgment about both machines and humans [55]. GPCC tools that promote engagement with young adults—a population with heightened social sensitivity— have profound developmental and societal implications. That GPCCs can provide social support across every life domain does not mean they should do so without constraint. We hold that works on improving CCs for young people must define the boundaries of their engagement—transparent about what they are, explicit about what they cannot replace, and constrained in how deeply they engage.

8 LIMITATIONS

This study has several limitations. First, our sample skews toward heavier ChatGPT users: the minimum-length inclusion filter likely excludes casual users whose support-seeking behaviors may differ. Although we recruited across 17 countries, the English-language filter necessarily excluded non-English-dominant users whose cultural contexts may shape support-seeking in ways our data cannot capture [58]. Second, our thematic analysis covers 250 manually coded conversations—approximately 2.6% of the dataset. Following reflexive thematic analysis principles [13, 14], our goal was interpretive depth rather than exhaustive data saturation[15]; the usage distributions we report characterize our coded sample, not population-level estimates.

Third, we analyze conversation logs without access to participants' motivations or outcomes. A user who copied entire homework assignments may have had different levels of learning engagement than our coding captures.. Future work should pair log analysis with retrospective accounts.

Fourth, our initial wave of data collection, which yielded the empirical data for this paper, did not include an explicit procedure for participants to sanitize their logs. We did not anticipate the depth of personal and sensitive self-disclosure we would encounter. While our subsequent data collection protocols have implemented active data sanitization, those data were not used for the present work.

Fifth, our dataset spans multiple ChatGPT model versions whose differing capabilities and safety behaviors we cannot disentangle from changes in user behavior over time.

Finally, although we frame our contributions around GPCCs broadly, our data are limited to ChatGPT. Other GPCCs—such as Gemini and Claude—differ in conversational design, safety constraints, and default response styles, and may produce different support patterns. Cross-platform comparison remains an important direction for future work.

9 CONCLUSION

Through mixed-methods analysis of 85 emerging adults' in-the-wild ChatGPT conversations—approximately 157,000 messages over three years—we documented ten support-seeking patterns across four life domains and five ChatGPT response patterns. Our findings reveal that tangible support, delivered primarily through Task Takeover, is the mechanism through which young adults form sustained relationships with GPCCs—creating pathways from instrumental use to emotional disclosure. This challenges the dichotomy between "task-oriented" and "social" CCs: for emerging adults, functional utility and emotional support are deeply intertwined. We synthesize these findings into a design space for developmentally supportive GPCCs, arguing that systems serving young people should shift from producing outputs on users' behalf toward scaffolding their thinking and action. That GPCCs can provide support across every life domain does not mean they should do so without constraint. Future work should examine how these response patterns shape user agency and long-term well-being as GPCCs continue to reshape the landscape of social support.

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